
Reinforcement Learning for Hospital Equipment Contract Coverage Optimization

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Abstract

1 Hospital equipment management at medical-technology companies such as Philips
2 relies on a complex portfolio of maintenance contracts, manufacturer warranties and
3 warranty extensions. Current decision-support tools remain largely descriptive and
4 provide limited anticipation of contractual risks. We cast the coverage-optimization
5 problem as a finite-horizon Markov decision process (MDP) on a single piece of
6 equipment and train a from-scratch Proximal Policy Optimization (PPO) agent
7 in a reproducible simulation environment. Our contribution is not algorithmic:
8 PPO, GAE and observation normalization are taken off-the-shelf. We contribute
9 a clean MDP formulation for a decision-making problem that, to our knowledge,
10 has not been cast as RL in the published literature, together with a statistically
11 rigorous evaluation protocol that includes synchronous value iteration as a tab-
12 ular Bellman reference. On 100 shared seeds, PPO attains 54.56 ± 3.93 (95%
13 CI [53.79, 55.33]), far above random (-241.44 ± 92.66), but below a threshold
14 heuristic (58.06 ± 3.93) and synchronous value iteration (58.08 ± 3.96), which
15 are statistically indistinguishable. The heuristic remains robust across its threshold
16 sweep, and a state-by-state tabulation of the saved PPO policy shows 95.2% action
17 agreement with VI over 65,880 reachable states.

18 1 Introduction

19 In large healthcare organizations such as Philips France, the management of medical equipment relies
20 on a complex portfolio of maintenance contracts, manufacturer warranties and warranty extensions.
21 This management is critical: it directly conditions the continuity of clinical services, the quality of
22 patient care, and the operational costs borne by the organization. The contractual portfolio evolves
23 continuously, driven by installations, renewals and expirations, which makes coverage mastery
24 particularly difficult at the scale of a large installed base.

25 Despite the growing availability of data from information systems, existing tools remain largely
26 descriptive and fragmented. Current dashboards offer partial visibility into contractual coverage, but
27 they neither anticipate risks associated with contract expirations nor prioritize actions in a proactive
28 manner. They also fail to provide explanations accessible to business users, limiting their ability to
29 interpret indicators, justify decisions, and act quickly in complex operational environments. The
30 result is a growing need for intelligent solutions capable of transforming data into actionable decisions
31 while delivering clear, contextualized justifications.

32 We argue that this problem is naturally expressed as sequential decision-making under uncertainty and
33 is therefore well suited to a reinforcement learning (RL) treatment. Specifically, we formalize it as a
34 *Markov decision process* (MDP) in which the state describes the set of managed pieces of equipment
35 and their current contractual posture (maintenance, manufacturer warranty, extension), the actions
36 correspond to coverage decisions (subscription, renewal, adjustment, prioritization), transitions model

the portfolio dynamics and contract expirations, and the reward balances coverage cost against the penalty associated with uncovered equipment and late interventions. In this framework, an RL agent learns a long-horizon coverage policy that anticipates expirations and prioritizes the actions with the greatest value for the organization.

Operationalizing this formulation raises several challenges. The environment is only partially observable and must be reconstructed from heterogeneous data sources; the action space combines discrete decisions over a large number of pieces of equipment; the cost of erroneous exploration in production precludes direct online learning, which motivates the use of simulators and offline evaluation protocols; finally, the learned policy must be interpretable to be adopted by the business teams responsible for contract management.

This work is positioned as an *applied case study*. We do not claim a methodological contribution: the PPO algorithm used here is standard and has been extensively studied in prior work Schulman et al. [2017], Andrychowicz et al. [2021]. Our contribution lies in the problem formulation, the reproducible experimental setup, and the rigorous head-to-head comparison with business-rule baselines in a domain—the coverage management of hospital equipment—that, to our knowledge, has not been cast as a reinforcement-learning problem in the published literature.

More specifically, the contributions of this work are:

- **Problem formalization.** We formalize single-equipment contract-coverage optimization as a finite-horizon Markov decision process, with a four-dimensional state aggregating coverage status, time to expiration, contract cost level and risk level, and a three-way action space covering the standard contractual operations (do nothing, renew, extend).
- **Reproducible simulator.** We develop a self-contained simulation environment reproducing the coverage, expiration and intervention dynamics under parametric control, with fully deterministic reproducibility via seeded initial states and failure processes.
- **Bellman reference.** We compute a greedy tabular optimum through discounted value iteration on an augmented finite MDP that mirrors the simulator’s transition logic in expectation. This certificate clarifies achievable reward under our modelling commitments and highlights when rule-based controllers are analytically aligned with DP.
- **Rigorous empirical comparison.** We train a Proximal Policy Optimization agent and compare it against a uniform random policy, a threshold-based heuristic spanning multiple threshold values, and the tabular VI rollout on 100 shared seeds (means, standard deviations, 95% confidence intervals, Welch tests).

Scope. We deliberately restrict the experiments to a single piece of equipment and a four-dimensional state so that every design choice can be stated, audited and reproduced. This is a term-paper-level study: we aim at providing a clean starting point and an honest baseline for future work on multi-equipment contractual portfolios.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 details the MDP formulation and the learning algorithm. Section 4 describes the experimental protocol and reports the empirical results. Section 5 discusses interpretations, limitations and perspectives, and Section 6 concludes.

2 Related Work

Our study lies at the intersection of three research strands: maintenance and service-contract optimization, reinforcement learning for sequential decision-making in industrial operations, and deep policy-gradient methods.

Maintenance and service-contract optimization. Service-contract design and maintenance scheduling have a long tradition in operations research, where they are typically formulated as integer programs, renewal-theoretic models or stochastic dynamic programs Jardine and Tsang [2006], Dekker [1996]. These approaches provide principled solutions under strong modelling assumptions but scale poorly to high-dimensional portfolios with stochastic failures and evolving contractual structures, and typically produce static rather than adaptive policies. In industrial practice,

87 simple threshold rules and segment-based heuristics dominate deployments Gits [1992], which moti-
 88 vates the search for data-driven alternatives that can capture the temporal feedback between coverage
 89 decisions and downstream costs.

90 **Reinforcement learning for industrial operations.** Reinforcement learning has emerged as a
 91 natural framework for sequential decision-making in industrial and financial operations, thanks to its
 92 ability to learn from interactions without requiring an explicit model of the environment Sutton and
 93 Barto [2018]. Recent applications include risk-sensitive trading agents trained with language-model-
 94 augmented rewards Benhenda [2025], deep RL for efficient liquidity provisioning in decentralized
 95 finance Xu and Brini [2025], and dynamic pricing and inventory management Boute et al. [2022].
 96 These works illustrate both the promise of learned decision-making under uncertainty and the
 97 recurring difficulties of reward specification, sample efficiency and safe deployment—difficulties that
 98 are also central to our setting.

99 **Reinforcement learning in healthcare operations.** Within healthcare, RL has been applied to
 100 treatment recommendation, dynamic treatment regimes and operational decision-making such as ICU
 101 ventilator weaning and resource allocation Yu et al. [2021]. Most of these applications emphasize
 102 safety, interpretability and offline learning from historical logs, all constraints that also apply when
 103 managing a fleet of medical devices. To the best of our knowledge, contract-coverage optimization
 104 for medical equipment has not yet been cast as an RL problem in the published literature, which
 105 motivates the present work.

106 **Policy-gradient methods.** At the algorithmic level, our approach builds on deep actor-critic
 107 methods. Proximal Policy Optimization Schulman et al. [2017] provides an empirically stable
 108 policy-gradient procedure through a clipped surrogate objective, which we couple with Generalized
 109 Advantage Estimation Schulman et al. [2016]. Running observation normalization and advantage nor-
 110 malization, which we also use, are standard components of modern PPO implementations Andrychow-
 111 icz et al. [2021]. Our contribution is not algorithmic but applicative: we design an MDP formulation,
 112 a simulation environment and an evaluation protocol tailored to contractual-coverage optimization,
 113 and we situate the learned policy against a random baseline and a strong rule-based heuristic.

114 3 Method

115 This section formalizes the contract-coverage optimization problem as a Markov decision process
 116 (MDP) and describes the reinforcement-learning algorithm we adopt, Proximal Policy Optimization
 117 (PPO). We restrict the formulation to the exact scope of our empirical study: a *single piece* of
 118 medical equipment followed over a finite daily horizon. This deliberate simplification makes the
 119 analysis tractable and keeps the theoretical description and the implementation in strict one-to-one
 120 correspondence; extensions to a portfolio of heterogeneous equipment are discussed in Section 5.

121 3.1 MDP formulation

122 We model the problem as a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, T)$, with daily time steps and
 123 horizon $T = 120$ days, tracking a single piece of equipment.

124 **State space.** The state at step t is a four-dimensional vector

$$s_t = (\text{covered}_t, \text{days_to_exp}_t, \text{cost_level}, \text{risk_level}),$$

125 where $\text{covered}_t \in \{0, 1\}$ indicates whether an active contract covers the equipment, days_to_exp_t
 126 is the remaining number of days before expiration of the current coverage, $\text{cost_level} \in \{0, 1, 2\}$
 127 discretizes the contractual cost (low, medium, high), and $\text{risk_level} \in \{0, 1, 2\}$ discretizes the
 128 equipment risk class (low, medium, high). The two discrete attributes are sampled at the start of each
 129 episode and remain constant throughout. All four components are normalized to the unit interval
 130 before being fed to the policy.

131 **Action space.** The action space is discrete with three elements:

$$\mathcal{A} = \{0 : \text{do nothing}, 1 : \text{renew contract}, 2 : \text{extend warranty}\}.$$

132 A *renewal* (action 1) resets the remaining coverage to a fixed duration (RENEWAL_DAYS = 30)
 133 and restores the covered status if the equipment was uncovered. A *warranty extension* (action 2)
 134 adds a fixed duration (EXTENSION_DAYS = 30, capped at MAX_COVERED_DAYS = 60) at
 135 a reduced cost, but requires an already-active contract; invoking it while uncovered incurs a small
 136 no-op penalty.

137 **Reward function.** The per-step reward aggregates six components, each chosen so that the total
 138 remains in a numerically stable range and induces a clear ordering between desirable and undesirable
 139 decisions:

$$r_t = r_t^{\text{coverage}} + r_t^{\text{timing}} + r_t^{\text{cost}} + r_t^{\text{expiry}} + r_t^{\text{early}} + r_t^{\text{failure}}.$$

140 The continuous coverage term is +COVERAGE_BONUS while covered and
 141 −UNCOVERED_PENALTY otherwise. The timing term rewards renewals or extensions
 142 issued close to expiration (+RENEWAL_CLOSE_BONUS when $\text{days_to_exp}_t \leq 5$) and
 143 penalizes early renewals (−RENEWAL_EARLY_PENALTY when $\text{days_to_exp}_t \geq 20$). The
 144 cost term is a contract cost proportional to cost_level (smaller for extensions than for full renewals).
 145 The expiration term is a one-shot penalty applied the exact step a contract expires without renewal.
 146 The failure term is a one-shot penalty whenever a failure occurs while the equipment is uncovered,
 147 sampled with probability proportional to $(1 + \text{risk_level})$.

148 **Transition dynamics.** At each step, the environment sequentially (i) applies the chosen action, (ii)
 149 decrements days_to_exp_t by one and triggers expiration if the counter reaches zero, (iii) samples a
 150 Bernoulli failure indicator with risk-conditional probability, and (iv) emits the scalar reward. The
 151 static attributes cost_level and risk_level are unchanged across a full episode, so the only stochastic
 152 transitions concern the coverage counter and the failure process. The dynamics are fully Markovian
 153 in the four-dimensional state.

154 3.2 Tabular Bellman optimum (value iteration)

155 For benchmarking and clarification of optimality notions, we also compute an exact greedy policy
 156 in a finite MDP that augments each physical state with the elapsed time index $\tau \in \{0, \dots, T-1\}$
 157 recorded by the simulator at each decision.

158 Because cost_level and risk_level are fixed throughout an episode, this augmented process is finite—
 159 at most $|\mathcal{S}_{\text{tab}}| \approx 65,880$ reachable tuples for our $T=120$, $D_{\text{max}}=60$ setting—so we solve it with
 160 synchronous discounted value iteration using the same $\gamma=0.99$ as PPO.

161 The associated greedy policy, denoted $VI(\text{exp.})$ in experiments, maximises the usual discounted
 162 criterion in *expectation* relative to this tabulation; Monte Carlo evaluation on the sampled simulator
 163 inherits Bernoulli noise, so confidence intervals remain non-degenerate.

164 3.3 Proximal Policy Optimization (PPO)

165 To approximate the MDP defined in Section 3.1, we use the *Proximal Policy Optimization* (PPO)
 166 algorithm Schulman et al. [2017], a policy-gradient method recognized for its empirical stability and
 167 implementation simplicity.

168 **Principle.** PPO belongs to the family of actor-critic methods. The algorithm jointly trains two
 169 components: a stochastic policy $\pi_\theta(a \mid s)$ that maps each state to a distribution over the three
 170 admissible actions, and a value function $V_\phi(s)$ that estimates the expected future return from state s .
 171 Both components are parameterized by neural networks sharing a common two-layer encoder (64
 172 units, tanh activations) with separate softmax policy and scalar value heads. They are learned from
 173 trajectories collected under the current policy. The advantage estimate, which measures the marginal
 174 gain of an action relative to the average value of the state, is obtained through Generalized Advantage
 175 Estimation (GAE) Schulman et al. [2016], with a discount $\gamma = 0.99$ and GAE coefficient $\lambda = 0.95$,
 176 providing a controllable trade-off between bias and variance.

177 **Clipped objective.** The specificity of PPO lies in its learning objective. Let $r_t(\theta) = \pi_\theta(a_t \mid$
 178 $s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$ denote the likelihood ratio between the current policy and the behavior policy that

179 produced the trajectory, and let $\hat{A}_t$ denote the normalized GAE advantage. PPO maximizes

$$\mathcal{L}^{\text{clip}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right],$$

180 with clip parameter $\epsilon = 0.2$. The value function is trained by minimizing the squared return-prediction
 181 error, and an entropy bonus is added to the global objective to encourage exploration and to prevent
 182 premature convergence to a deterministic policy.

183 **Implementation details.** Observations are normalized online through a running estimate of their
 184 mean and variance (Welford’s algorithm) and clipped to $[-5, 5]$ before being fed to the network.
 185 Advantages are normalized per batch. At each iteration, a full episode is collected under the current
 186 policy, and the policy–value pair is updated for four epochs over mini-batches of size 64. Training is
 187 conducted entirely offline on the simulator, which enables intensive exploration without operational
 188 risk and yields a reproducible evaluation of the learned policy prior to any deployment.

189 4 Experimental Setup

190 This section describes the simulated environment, the baselines, the training setup and the evaluation
 191 protocol, and reports the empirical results with proper statistical summaries.

192 4.1 Dataset and simulation environment

193 Direct online access to Philips France’s contractual inventory is subject to confidentiality and op-
 194 erational constraints, so we do not conduct training on the production systems. Instead, we build
 195 a *simulation environment* whose parameter distributions are chosen to reproduce the qualitative
 196 characteristics of a hospital-equipment installed base: coverage levels, contract durations, contractual
 197 costs, and equipment risk levels. This environment plays the role of both dataset and interaction
 198 simulator, which guarantees full experimental reproducibility and avoids any exposure of sensitive
 199 data.

200 Each episode represents a single piece of medical equipment tracked over a finite horizon of $T = 120$
 201 daily time steps. At reset, the environment randomly draws: a remaining time before current coverage
 202 expiration within $[5, 30]$ days, a contract-cost level with three modalities (low, medium, high), and a
 203 risk level with three modalities (low, medium, high). The equipment is initially covered.

204 4.2 Baseline policies

205 We compare the PPO agent with three complementary baselines. The first is a uniformly random
 206 policy over the three actions, which sets a lower performance bound. The second is a rule-based
 207 heuristic reflecting operational practice, with a nominal threshold of five days. The third is the greedy
 208 policy computed by synchronous value iteration on the expected-reward tabular MDP of Section 3.2,
 209 denoted VI (*exp.*): discounted Bellman-optimal in expectation, rolled out with the identical simulator
 210 as every other baseline. Welch’s two-sided test on per-episode rewards between VI and heuristic
 211 returns $t \approx 0.04$, $p \approx 0.97$ ($n=100$ shared seeds), which is consistent with the hypothesis that
 212 threshold rules saturate the Bellman-optimal frontier in expectation here.

213 4.3 Architecture and training setup

214 The policy and value function share a two-layer fully connected encoder of width 64 with tanh
 215 activations, followed by two linear heads: a softmax head over the three actions and a scalar value
 216 head. Observations are normalized online through a running estimate of the mean and variance (using
 217 Welford’s algorithm), then clipped to the interval $[-5, 5]$ before being fed to the network.

218 Training uses PPO with Generalized Advantage Estimation (GAE). At each iteration, a full episode is
 219 collected under the current policy, the advantage is computed, and the policy-value pair is updated
 220 for four passes over mini-batches of size 64. Advantages are normalized per batch. The objective
 221 combines PPO’s clipped loss, a squared value loss and an entropy bonus encouraging exploration.
 222 Hyperparameters are summarized in Table 1.

Hyperparameter	Value
Number of episodes	5,000
Episode horizon	120 days
Learning rate	3×10^{-4}
Discount factor	0.99
GAE parameter	0.95
Clip range	0.2
Optimization epochs	4
Mini-batch size	64
Hidden dimension	64
Entropy coefficient	0.02
Value coefficient	0.5
Observation clipping	$[-5, 5]$

Table 1: Training hyperparameters.

223 4.4 Evaluation protocol

224 Every 100 training episodes, the current policy is evaluated in deterministic mode (argmax action
 225 selection) over 20 episodes sharing fixed simulation seeds. The best policy in terms of evaluation
 226 reward is retained and restored at the end of training.

227 The *final* comparison with the baseline policies is conducted on a separate and larger pool of 100
 228 simulation seeds, shared across every policy, so that they face identical contract and failure trajectories.
 229 This procedure neutralizes the stochastic variance of the environment and enables a statistically sound
 230 comparison between random, heuristic, VI and learned controllers. We report, for every metric
 231 and every policy, the sample mean, the sample standard deviation and a two-sided 95% confidence
 232 interval on the mean computed with Student’s t -distribution ($df = 99$).

233 4.5 Results

234 The performance on 100 shared evaluation seeds is reported in Table 2.

Metric	Random	Heuristic	PPO	VI (exp.)
Cumulative reward (mean $\pm$ SD)	-241.44 ± 92.66	58.06 ± 3.93	54.56 ± 3.93	58.08 ± 3.96
95% CI	$[-259.60, -223.28]$	$[57.29, 58.84]$	$[53.79, 55.33]$	$[57.31, 58.86]$

Table 2: Performance comparison over 100 evaluation episodes with shared seeds. Each cell reports mean $\pm$ SD on the first line and the 95% confidence interval on the second (Student’s t , $df = 99$).

235 **Main finding.** The heuristic attains a mean reward of 58.06 and PPO attains 54.56; the gap is
 236 far outside the respective 95% confidence intervals. Welch’s test on per-episode rewards yields
 237 $t = -6.31$, $p < 0.0001$ ($df \approx 198$). The tabular Bellman policy (VI) attains 58.08 ± 3.96 ,
 238 statistically indistinguishable from the heuristic (Welch $t \approx 0.04$, $p \approx 0.97$), so the latter should be
 239 understood as *co-optimal* with the exact finite-horizon tabulation, not as an arbitrary engineering
 240 baseline. PPO’s gap is therefore a gap to the Bellman-optimal frontier of this toy MDP, driven by
 241 function approximation and finite training rather than by a weak reference policy.

242 **A counter-intuitive but well-explained random baseline.** The random policy obtains a strongly
 243 negative reward despite recording zero uncovered steps and zero uncovered failures in every evaluated
 244 episode. This apparently counter-intuitive result stems from the structure of the action space: by
 245 drawing uniformly among the three decisions, two of which—renew and extend—preserve or
 246 restore coverage, the random agent mechanically keeps contracts active but accumulates prohibitive
 247 contractual costs. The absence of a strategy thus manifests as expensive over-coverage rather than
 248 under-coverage.

249 **PPO learns a competent but sub-optimal policy.** PPO reaches a clearly positive mean reward
 250 and demonstrates that a standard policy-gradient procedure is able to learn, from interactions with

the simulator alone, perfect coverage scheduling on these seeds while still paying renewal costs imperfectly relative to the threshold certificate. It is not competitive with the heuristic in mean reward, despite the elongated 5,000-episode budget. The residual mean gap is discussed in Section 5; in essence, VI shows the MDP’s parsimonious frontier is encodeable by a one-parameter rule whereas PPO must recover it indirectly from noisy gradient feedback.

4.6 Sensitivity analysis: heuristic threshold

A natural concern is that the strong performance of the heuristic might come from a specific, favourable choice of its renewal threshold. Table 3 reports the mean reward of the heuristic for five values of the threshold, evaluated on the same 100 shared seeds.

Threshold (days)	3	5	7	10	15
Mean reward $\pm$ SD	58.06 $\pm$ 3.93	58.06 $\pm$ 3.93	49.98 $\pm$ 4.00	49.74 $\pm$ 4.23	49.14 $\pm$ 4.62
95% CI	[57.29, 58.84]	[57.29, 58.84]	[49.20, 50.76]	[48.91, 50.57]	[48.24, 50.04]

Table 3: Heuristic renewal-threshold sensitivity on the same 100 shared seeds as Table 2.

The heuristic’s peak is flat: thresholds of 3 and 5 produce exactly the same mean reward (58.06). The heuristic is robust across its operating range: the largest observed loss is only 8.9 reward units between the peak and threshold 15. At 54.56 ± 3.93 , PPO improves on the loosest heuristic instantiation while remaining materially below VI and the rule’s peak regime.

4.7 Interpretability via Bellman tabulation

To localise where the PPO greedy policy departs from the Bellman certificate, we tabulate its action choice on every cell of the augmented finite MDP: state (covered, days_to_exp, cost_level, risk_level, step_index), with $|S_{\text{tab}}| = 65,880$ reachable cells.

The headline numbers are an action-agreement of 95.2% with the VI optimum and 95.6% with the threshold heuristic. Stratifying by coverage status further isolates the failure mode: on covered cells (64,800 states), PPO agrees with VI on 95.5% and with the heuristic on 95.9%; nearly all disagreement comes from PPO extending instead of doing nothing. On uncovered cells (1,080 states), agreement drops to 77.8% with both references.

4.8 Scope of the empirical claims

The results in this section are obtained in a deliberately simplified simulation environment whose distributions are controlled by a handful of parameters. They therefore do not directly entail the performance achievable in production, where additional factors—portfolio heterogeneity, global budget constraints, partial observability, non-stationary dynamics—that are absent from the present framework come into play. Within this scope, the results motivate a sharper claim than “PPO loses against a heuristic”: VI shows the heuristic coincides—up to Monte Carlo noise—with an exact Bellman policy, whereas PPO remains about 3.5 reward units behind on average ($p \ll 10^{-4}$).

5 Discussion

The results reported in Section 4.5 place reinforcement learning significantly above the random policy but also significantly below both the heuristic rule and the tabular VI benchmark, with Welch tests against PPO at $p < 10^{-4}$ whereas heuristic and VI rewards are statistically inseparable ($p \approx 0.97$). This section interrogates how to read these contrasts and sketches responsible extensions.

5.1 Positioning of the contribution

Before discussing the empirical gap in detail, it is useful to restate what this paper does and does not claim. We do not contribute a new RL algorithm: every component of our pipeline (PPO, GAE, observation normalization, advantage normalization, entropy bonus) is taken off-the-shelf from prior

work Schulman et al. [2017, 2016], Andrychowicz et al. [2021]. The contribution is applicative and methodological in a narrower sense: we isolate a business decision-making problem whose RL formulation has not, to our knowledge, been published; we build the smallest non-trivial simulator capable of carrying that decision; we derive a tabular Bellman reference via synchronous value iteration; and we compare PPO against uniform random rolls, heuristic sweeps, and the VI roll-out with shared seeds, confidence intervals and Welch tests.

5.2 Why PPO outperforms the random policy

The gap between PPO and the random policy does not merely reflect a difference in immediate expected reward; it stems from the fact that the agent learns to *structure* its decisions around the state. Three mechanisms drive this behavior.

First, the agent identifies a *semantics* of coverage status: it learns that renewing is of little value—or even detrimental—while coverage is still largely valid, and that it becomes critical as expiration approaches. This semantics is not provided upfront: it is extracted exclusively from the reward signal, through observed trajectories.

Second, the agent learns a *temporal credit assignment*: the GAE estimator propagates the penalties of expiration and of uncovered failure back to the decisions that made them possible several steps earlier. The random policy, which does not adapt to the current state, cannot benefit from this information and accumulates penalties mechanically.

Third, the agent *conditions* its decisions on the full state, which lets it implicitly weigh the cost of a renewal against the contract-cost level and the risk level. The random policy treats all situations interchangeably and is necessarily penalized by this lack of specificity.

5.3 Why the heuristic policy can still win

The gap between PPO and the threshold heuristic is best read through the lens of *co-optimal planning*. Value iteration on the augmented tabular MDP yields roll-out statistics that Welch’s test cannot distinguish from the heuristic, so the “business rule” is not an arbitrary straw man: it tracks the Bellman reference up to Monte Carlo noise. Neural PPO therefore trails an analytical certificate, not merely tribal knowledge.

The sensitivity analysis in Section 4.6 adds engineering colour: the heuristic sits on a flat optimum for low thresholds, stays robust until the chosen deadline drifts excessively late, and attains peaks that PPO has not matched despite outperforming its loosest instantiations. Production teams may value that robustness, while researchers should treat it as evidence that parsimonious policies exhaust the toy model’s headroom until couplings and partial observability enter the story.

5.4 Limitations of the current setup

Several limitations bound the scope of the reported conclusions.

Simulator dependence. Training and evaluation are conducted exclusively in a simulated environment. Distributions over contract durations, cost levels and risk, as well as failure probabilities, are defined by fixed parameters that have not been matched against Philips France’s real installed base. Any bias between these distributions and reality will mechanically propagate to the learned policy.

Single-equipment framing. The MDP models a single piece of equipment at a time, without interactions between contracts. This choice simplifies analysis and reproducibility but elides budget couplings, volume constraints and portfolio effects that lie at the heart of the original operational problem.

Sensitivity to reward shaping. The reward function blends manually chosen coefficients. An exhaustive sensitivity programme over coupled coefficients remains a natural extension.

Limited interpretability. The learned policy is represented by a neural network whose decisions are not natively explainable. As a first step, Section 4.7 tabulates PPO’s greedy action on every reachable cell of the augmented finite MDP and reports the per-state agreement with the VI optimum

and with the threshold heuristic. This diagnostic does not replace a fully distillable surrogate but it pinpoints where the policy departs from the Bellman certificate.

Governance and human oversight. High-stakes renewals that affect patient-ready equipment should remain under human-in-the-loop approval, with immutable audit logs and periodic reviews of simulator parameters and reward weights to surface institutional or financial bias before automation scales.

No hard operational constraints. Contractual budget, exclusivities, renewal eligibility windows and contractual obligations do not appear explicitly in the MDP. In a production deployment, such constraints must be *guaranteed*, not merely encouraged by the reward.

5.5 Paths for improvement

The limitations above suggest several concrete directions.

Extension to the full installed base. Moving from the single-equipment framing to a heterogeneous portfolio exposes budget couplings and risk asymmetries that naïvely blow up tabular planning. Factorised critics, constrained optimisation and amortised RL then address regimes where threshold heuristics and brute-force VI lose traction together.

Offline learning from historical data. Once available, the operational traces of Philips France may serve as the basis for offline RL, avoiding any interaction with the production system. Algorithms such as Conservative Q-Learning or Implicit Q-Learning allow exploiting such data while controlling extrapolation risk.

Constrained learning. Budgetary and contractual constraints may be enforced explicitly, either via action masking (prohibition at the softmax level of infeasible actions) or through constrained-RL algorithms such as Constrained PPO and Lagrangian methods.

Robustness and uncertainty quantification. Training over an ensemble of parametrically diverse simulators, the use of Bayesian policies, and joint perturbation grids would strengthen out-of-distribution guarantees.

Interpretability. Beyond the state-by-state agreement diagnostic, several additional levers may reduce policy opacity: distillation of the learned policy into a decision tree or a set of rules, analysis of input-feature importance, or hybrid formulations combining an interpretable rule with a learned residual.

Comparison with stochastic optimization. Finally, confronting PPO with classical optimization methods—multi-stage stochastic programming, rolling-horizon integer programs—would complement the study with a comparison against a global policy obtained directly from an explicit model of the problem.

These directions converge toward a more demanding experimental framework in which the simplifying assumptions of the present work are lifted one by one. They form the core of the research program outlined in Section 6.

6 Conclusion

This work addresses the optimization of contractual coverage for medical equipment, with the operational goals of reducing the fraction of uncovered equipment and improving contractual decision-making. The problem is formalized as a finite-horizon Markov decision process on a single piece of equipment and solved with the PPO algorithm, trained entirely in a reproducible simulator built in the absence of direct access to Philips France’s operational data. The contribution is applicative: we do not propose a new algorithm, but we provide a clean MDP formulation, a reproducible training pipeline, and a statistically grounded comparison with three baselines: uniform random actions, a

threshold heuristic whose parameter is swept, and a greedy policy from discounted value iteration on the tabulated expected MDP.

On 100 evaluation episodes drawn from a shared seed pool, the PPO agent reaches a mean reward of 54.56 ± 3.93 (95% confidence interval $[53.79, 55.33]$), far above the random policy (-241.44 ± 92.66) yet still below both the heuristic (58.06 ± 3.93) and VI (58.08 ± 3.96), with Welch tests against PPO at $p < 10^{-4}$ for either co-optimal reference, while heuristic and VI means are statistically indistinguishable ($p \approx 0.97$). Threshold sweeps trace a flat peak at 58.06 before decaying to 49.14 at threshold 15: PPO therefore sits between the loosest heuristic and the Bellman-aligned plateau.

A state-by-state interpretability tabulation localises the disagreement: PPO matches VI on 95.2% of 65,880 reachable states, with residual mistakes concentrated in covered-state *do nothing* versus *extend* trade-offs. Taken together, these observations show that PPO recovers non-trivial behaviour from reward alone but does not match any certificate of Bellman optimality available in this toy model. Reinforcement learning’s comparative advantage is expected only once portfolios, partial observability and budget couplings break the parsimony that tabulation already exhausts here.

These results rely on a deliberately simplified simulator, a four-dimensional state representation and a training budget of 5,000 episodes, all of which bound the scope of the conclusions. Natural extensions include the exploitation of real operational data to enrich the state representation, the extension of the framework to heterogeneous portfolios subject to global budget constraints, a sensitivity study of PPO itself to changes in reward weights and environmental parameters, and ultimately a supervised deployment in production.

The presented approach demonstrates the viability of driving contractual decisions through reinforcement learning and opens the way to a gradual, empirically justified adoption of learned policies complementary to existing business rules.

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